

LAB Program-3A.

WAP to stimulate the working of a queue of integers using an array. Provide the following operations: Insert, Delete, Display. The program should print appropriate messages for queue empty and queue overflow conditions.

```
#include <stdio.h>
```

```
#define N5
```

```
int queue[N];
```

```
int front
```

PSEUDO CODE:

```
void enqueue(x){  
    if (rear == N-1){  
        printf("Overflow");  
    }  
    else if (front == -1 && rear == -1){  
        printf("Queue is empty");  
        front = rear = 0;  
    }  
    else queue[rear] = x;  
}
```

```
    else {  
        rear++;  
        queue[rear] = x;  
    }  
}
```

```
void dequeue(){  
    if (front == -1 && rear == -1){  
        printf("Queue is empty");  
    }  
}
```

```

else if (front == rear) {
    front = rear - 1;
}
else {
    printf("Delete ele = %d", queue[front]);
}

void display () {
    if (front == -1 && rear == -1) {
        printf("Queue is empty");
    }
    else {
        for (i = front; i <= rear; i++) {
            printf("%d ", queue[i]);
        }
    }
}

```

~~AB~~

CODE :

```

#include <stdio.h>
#define N 5
int queue[N];
int front = -1;
int rear = -1;

void enqueue () {
    int x;
    printf("Enter a number: ");
    scanf("%d", &x);
    if (rear == N) {
        printf("Queue Overflow");
    }
}

```



```

else if (front == -1 && rear == -1) {
    front = rear = 0;
    queue[rear] = x;
}
else {
    rear++;
    queue[rear] = x;
}
}

int dequeue() {
    if (front == -1 && rear == -1) {
        printf("Queue is empty\n");
    }
    else if (front == rear) {
        front = rear = -1;
    }
    else {
        printf("Deleted element = %d\n", queue[front]);
        front++;
    }
}

```

```

int display() {
    int i;
    if (front == -1 && rear == -1) {
        printf("Queue is empty\n");
    }
    else {
        printf("Queue:\n");
        for (i = front; i <= rear; i++) {
            printf("%d\n", queue[i]);
        }
    }
}

```

```

int main () {
    int ch;
    do {
        printf("Enter your choice '1' for Insertion,
               '2' for Deletion & '3' for Display:");
        scanf("%d", &ch);
        switch (ch) {
            case 1:
                enqueue();
                break;
            case 2:
                dequeue();
                break;
            case 3:
                display();
                break;
            default:
                printf("Invalid Input");
        }
    } while (ch != -1);
}

```

Output:

Enter your choice '1' for Insertion, '2' for Deletion
'3' for Display: 2

Enter a number: 10

Enter your choice '1' for Insertion, '2' for Deletion
'3' for Display: 1

Enter a number: 11

Enter your choice '1' for Insertion, '2' for Deletion
'3' for Display: 2

Deleted element = 10

Enter your choice '1' for Insertion, '2' for Deletion
& '3' for Display: 1

Enter a number: 12

Enter your choice '1' for Insertion, '2' for Deletion
& '3' for Display: 3

Queue:

11

12.

~~13~~
13/10/24